

Nonnegative and positive factorizations

The maximum cp-rank in order six

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Challenging because sharp cp-rank control is still missing for the remaining order-six boundary configurations; community importance is the size of exact completely positive certificates.

Area: size of nonnegative symmetric factorizations

Context and notation

A symmetric matrix A is **completely positive** if $A = BB^T$ for an entrywise nonnegative real matrix B with finitely many columns. Let $\mathcal{CP}_n$ be the cone of these matrices of order n . Its **cp-rank**, $\text{cpr}(A)$, is the minimum number of columns in such a factor, with $\text{cpr}(0) = 0$. Boundaries and interiors use the usual Euclidean topology on the vector space of real symmetric matrices.

Problem statement

Is every real completely positive matrix of order six a sum of at most nine nonnegative rank-one matrices? Equivalently, is

$$\max_{A \in \mathcal{CP}_6} \text{cpr}(A) = 9,$$

or, equivalently, does each $A \in \mathcal{CP}_6$ admit $A = BB^T$ with $B \in \mathbb{R}_{\geq 0}^{6 \times 9}$, allowing zero columns?

The lower bound nine is established. Order six is the unresolved case of the original Drew–Johnson–Loewy bound: order five satisfies it, whereas counterexamples exist in higher orders. It is included as the specific remaining case identified in the literature, rather than as one item in a dimension-by-dimension list.

A known reduction places a matrix attaining the order-six maximum on the boundary of $\mathcal{CP}_6$, with full ordinary rank and at least one zero entry. This reduction does not establish the nine-column bound.

References

Abraham Berman, Mirjam Dür, and Naomi Shaked-Monderer, *Open problems in the theory of completely positive and copositive matrices*, Electronic Journal of Linear Algebra **29** (2015), 46–58, §4.2, p. 53. Naomi Shaked-Monderer, *On the DJL conjecture for order 6, corrected arXiv v3*, 2017-06-01; journal version in Operators and Matrices **11** (2017), 71–88, Theorem 1.1 and Corollary 3.2.

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Rechecked [Shaked-Monderer’s corrected v3, Theorem 1.1 and Corollary 3.2](#), and searched for later order-six DJL resolutions. The nine-term bound is proved on the boundary classes orthogonal to exceptional extremal copositive matrices, including positive nonsingular boundary matrices. Remaining maximizers can have a zero entry; this reduction is not a proof for all order-six matrices. [Behling et al. \(2026\), §5.6](#), reproduces counterexamples in orders seven and twelve, not six. No full resolution was located, and the exact open-status source remains historical.